Data Structures and Algorithms Laboratory

Assignment# 2 (2 Weeks)

1. Write a C program to build a complete binary tree using linked list. Take the elements from a file, which is to be given as the command line argument.
2. Write C programs for printing (i) inorder, (ii) postorder, and (iii) preorder traversals of the tree, generated by problem no. 1.
3. Write a C program to reconstruct the tree from its inorder and preorder traversals.
4. Write a C program to build a max heap. Take the elements from a file, which is to be given as the command line argument.
5. Write a C program to build a min heap. Take the elements from a file, which is to be given as the command line argument.
6. Write insertion and deletion functions in C for a max heap represented as a lined binary tree. Assume that each node has a parent field as well as the usual left child, right child and data fields.
7. Write a C program to build a binary search tree (BST) using linked list. Take the elements from a file, which is to be given as the command line argument.
8. Write a C program to print nodes of a BST in descending order.
9. Write a C program to build an AVL tree using linked list. Take the elements from a file, which is to be given as the command line argument.
10. Write a C program to delete a node in an AVL tree, as generated in problem no. 9.